Cross-SPECL: Cross-Scene Hyperspectral Image Classification via Spectral Stability Exploiting and Causal Learning

Xi Chen, Maojun Zhang, Yuxiang Liu, Chen Chen, and Shen Yan

***Abstract*—Single-source domain generalization (DG) presents a formidable challenge in cross-scene hyperspectral image (HSI) classification, as models trained on a single domain often struggle to generalize due to spectral shifts. Prevailing approaches predominantly concentrate on expanding the source distribution through data augmentation or adversarial training techniques. However, such strategies often neglect a critical pitfall. In the absence of explicit constraints, models are prone to overfitting domain-specific spurious correlations rather than capturing intrinsic invariant semantics. Moreover, the reliance on adversarial training frequently introduces instability and convergence difficulties. To address these limitations, this article proposes Cross-SPECL, a unified nonadversarial framework that transitions the paradigm to a structured "Stabilize-then-Disentangle" process. Specifically, the proposed framework first introduces a spectral patch low-frequency transformation (SPLT) network as a feature stabilizer. This module utilizes a dual-branch architecture wherein an auxiliary branch applies controlled perturbations to low-frequency components, thereby compelling the main branch to extract features resilient to style variations. To further consolidate this robustness, a synergistic feature stabilization (SFS) mechanism integrates supervised contrastive learning (SCL) and prototype contrastive learning (PCL) to enforce invariance to these perturbations and construct highly representative class-conditional prototypes. Building upon this foundation, the framework learns a domain-agnostic directed acyclic graph (DAG) derived from these prototypes to explicitly uncover the causal structure. This process generates a causal mask that effectively prunes domain-specific spurious features, ensuring robust generalization. Extensive experimental evaluations on three cross-scene HSI classification benchmarks demonstrate that Cross-SPECL yields significant performance improvements over state-of-the-art methods.**

***Index Terms*—Causal inference, directed acyclic graph (DAG), hyperspectral image (HSI), prototype contrastive learning (PCL), supervised contrastive learning (SCL).**

I. Introduction

HYPERSPECTRAL image (HSI) classification involves categorizing each pixel in an HSI into predefined classes based on their spectral signatures. Compared to traditional RGB and multispectral images, HSIs capture information by measuring the radiative flux reflected from the ground and atmosphere across hundreds of narrow, continuous spectral intervals within the visible and infrared reflective regions [1]. This approach provides rich spatial and spectral information for each pixel, enabling more precise identification of materials and objects. This technique benefits a wide range of fields, including mineral exploration [2], ecological conservation [3], agricultural management [4], and wetland monitoring [5].

Recently, hyperspectral foundational models like HyperSIGMA [6] and HyperFree [7], alongside advanced architectures such as SSEFN [8] and MMD-MLP [9], have significantly advanced spectral representation learning. Nevertheless, these methods predominantly operate under the assumption that training and testing data are independent and identically distributed (i.i.d.). This assumption is frequently violated in practical applications due to the spectral variability of the same land cover class across different scenes, a phenomenon commonly referred to as domain shift. Originating from a combination of temporal and geographical variations, as well as acquisition-specific factors such as sensor nonlinearity, seasons, and weather conditions [10], domain shift poses a critical obstacle. Consequently, classifiers trained exclusively on data from a source domain (SD) often struggle to generalize and exhibit significant performance degradation when applied to an unseen target domain (TD) [11].

Domain adaptation (DA) techniques [12] were widely employed to mitigate model performance degradation in cross-scene HSI classification. These methods typically learned cross-domain shared knowledge [13] by aligning feature distributions between the SD and TD using techniques like supervised contrastive learning (SCL, e.g., SCLUDA [14]), class-aligned generative adversarial adaptation (e.g., CCGDA [15]), or graph optimization with maximum mean discrepancy (e.g., topological structure and semantic information transfer network (TSTnet) [16]). Although effective, their requirement for access to TD data during training rendered the learned models target-specific. Consequently, model retraining was necessary when encountering new, unseen scenes, limiting their practical applicability.

Recently, domain generalization (DG) has emerged as a promising alternative, aiming to learn cross-domain invariant information solely from SD data to generalize to any unseen TD. The prevailing strategy in DG was data diversification, which sought to extend the SD distribution to cover

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The authors are with the College of Systems Engineering, National University of Defense Technology, Changsha, Hunan 410005, China (e-mail: xi_chen@nudt.edu.cn; mjzhang@nudt.edu.cn; liuyuxiang17@nudt.edu.cn; chenchen16@nudt.edu.cn; yanshen12@nudt.edu.cn).

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potential domain shifts. For instance, single-source domain extension network (SDEnet) [17] and LLURnet [18] employed contrastive adversarial learning or randomization techniques to generate augmented domains with diverse spatial–spectral variations. Other approaches utilized generative networks to simulate target-like biases [19] or synthesize diverse fused features [20]. While these methods showed promise, they suffered from two critical limitations. First, simply diversifying data did not guarantee that the model ignored nontransferable features; instead, the model might overfit to the expanded noise or spurious correlations [21]. Second, many of these methods relied on adversarial training, which is notoriously unstable and prone to model collapse.

To address these fundamental challenges, we propose Cross-SPECL, a novel framework that shifts the DG paradigm from simple diversification to a structured process of "Stabilize-then-Disentangle." Rather than relying on unstable adversarial generation or blind data augmentation, we argue that robust generalization requires a systematic approach: first, constructing a feature space that is inherently resilient to spectral style variations, and second, explicitly identifying and pruning spurious correlations within this stable space.

Cross-SPECL orchestrates a synergistic interaction among feature extraction, contrastive alignment, and causal discovery through a cohesive, nonadversarial workflow. First, to implement the "Stabilize" phase, we construct a spectral patch low-frequency transformation (SPLT) network. This network employs a dual-branch architecture incorporating an auxiliary branch that injects nonlearnable perturbations into the low-frequency components of the intermediate features. By leveraging SCL to enforce consistency between the main branch and these perturbed auxiliary views, we compel the encoder to disregard domain-specific spectral variations and focus on robust semantic representations. Second, to further consolidate the feature space, we employ prototype contrastive learning (PCL) to align features with global class centers. This process yields highly representative class-conditional prototypes that accurately capture the stable distribution of the SD. Finally, in the "Disentangle" phase, we utilize these stabilized prototypes to learn a domain-agnostic directed acyclic graph (DAG). This DAG uncovers the underlying causal structure of the data and generates a causal mask. By explicitly filtering out spurious correlations based on this mask, we ensure that the final prediction relies solely on domain-invariant causal factors.

The main contributions of this article are summarized as follows.

1) We propose Cross-SPECL, a unified nonadversarial DG framework that enhances generalization through a "Stabilize-then-Disentangle" paradigm, synergizing feature stabilization with causal discovery to explicitly mine domain-invariant features.
2) We design an SPLT network that utilizes a low-frequency perturbation (LFP) auxiliary branch. Combined with synergistic feature stabilization (SFS), this mechanism enforces robustness against spectral style shifts and yields highly representative class prototypes.
3) We introduce a domain-agnostic DAG learning module that operates on stabilized prototypes. This module discovers the underlying causal structure to generate a causal mask and enforces causal prototype alignment (CPA) to effectively filter out spurious features for robust cross-scene classification.
4) Extensive experiments on three cross-scene HSI classification benchmarks demonstrate that Cross-SPECL significantly outperforms state-of-the-art methods, validating the effectiveness of the proposed synergistic design.

The remainder of this article is organized as follows. Section II provides a review of related work. Section III introduces the theoretical preliminaries. Section IV details the proposed Cross-SPECL framework. Section V presents extensive experimental results and analysis, demonstrating the effectiveness of the proposed method. Finally, Section VI concludes this article.

II. Related Work

A. Causal Inference for DG

Causal inference has recently garnered significant attention in the computer vision community as a robust alternative to traditional statistical correlation-based learning. While standard correlation learning is susceptible to spurious associations, causal methods aim to uncover the underlying invariant mechanisms governing data generation [22]. In the context of DG, recent approaches attempted to integrate structural causal models (SCMs) to disentangle causal factors from domain-specific confounders.

Several innovative strategies were proposed to learn causal structures. For instance, the dynamic causal space method [23] integrated graph structure directly into the scoring function to facilitate DAG learning. To address scalability, scalable Bayesian causal discovery frameworks [24] were developed to sample DAGs from posterior distributions without explicit regularization. Furthermore, reinforcement learning-based approaches, such as those utilizing sequential graphs [25], modeled the topological order of DAGs to approximate posterior distributions efficiently. Despite these advancements, applying general causal discovery algorithms to the high-dimensional and complex spectral–spatial data of HSIs remained a challenge, particularly in single-source scenarios where observational diversity is severely limited. Unlike these general methods, our approach specifically tailors the causal discovery process to the spectral stability of HSIs via prototype-based learning.

B. Contrastive Learning

Contrastive learning emerged as a dominant paradigm in representation learning, demonstrating remarkable success in extracting discriminative features by maximizing the similarity between positive pairs while minimizing it for negative pairs [26]. Seminal works laid the foundation for this field: SimCLR [27] highlighted the necessity of strong data augmentation and large batch sizes for learning robust representations. To decouple the dependency on batch size, MoCo [28] introduced a dynamic dictionary with a momentum-updated encoder to maintain a consistent queue of negative samples. Subsequent advancements sought to simplify the training

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objective. BYOL [29] demonstrated that high-quality representations could be learned without negative pairs by relying on asymmetric network updates and prediction tasks. Similarly, SwAV [30] incorporated online clustering to enforce consistency between different augmented views of the same image. Barlow Twins [31] proposed a redundancy-reduction objective by cross-correlating the outputs of twin networks. In the realm of HSI classification, contrastive learning was increasingly employed to align features across domains. In our framework, we synergize these principles with low-frequency spectral perturbations to enforce feature stability, thereby providing a robust initialization for subsequent causal discovery.

C. Frequency-Domain Learning for DG

Frequency-domain analysis has been extensively explored in digital signal processing and recently adapted for computer vision tasks to address domain shift problems. The theoretical foundation lies in the Fourier Transform, which decomposes an image into amplitude and phase components. A growing body of research [32] suggested that the amplitude spectrum, particularly its low-frequency components, predominantly encodes low-level statistics (e.g., illumination, color, and smooth style variations). In contrast, the phase spectrum and high-frequency components preserve high-level semantic information, such as object structures and edges, which are critical for robust classification.

Based on this theory, recent representative studies proposed manipulating frequency components to disentangle domain-invariant semantics from domain-specific styles. For instance, ALOFT [33] introduced a dynamic low-frequency transform within a lightweight multilayer perceptron (MLP) architecture to capture global dependencies, positing that low-frequency components contained valuable global structural information, though they also carried style biases. To mitigate the risk of overfitting to these source-specific styles, PASTA [32] proposed a structured perturbation strategy that perturbed amplitude spectra, altering style information while preserving spatial semantics. Similarly, SCSD [34] enhanced generalization by enforcing semantic consistency while diversifying styles, which essentially manipulated the style-correlated low-frequency information. Conversely, HMCA [35] argued for focusing on high-frequency components to learn domain-invariant features, employing Fourier transform-based data augmentation to simulate domain shifts.

III. Preliminaries

Before detailing the proposed framework, we introduce the theoretical foundations of causal DG and the mathematical principles underlying differentiable causal structure learning, specifically focusing on SCMs and the dual role of the matrix exponential in ensuring acyclicity and quantifying causal effects.

A. Causal Perspective on DG

Domain shift stems from inherent variations in data-generating processes across diverse environments. From a causal perspective [22], observational data can be modeled as being generated from a set of latent factors: domain-invariant causal features ($\mathbf{z}_c$), domain-specific style features ($\mathbf{z}_s$), and spurious features ($\mathbf{z}_f$). In a single-source scenario, the data generation process is governed by functions $g_f(\cdot)$, $g_s(\cdot)$, and $g_y(\cdot)$

$$\mathbf{z}_f = g_f(y) + \boldsymbol{\epsilon}_f, \quad \mathbf{z}_s = g_s(\boldsymbol{\epsilon}_s), \quad y = g_y(\mathbf{z}_c) + \epsilon_y \tag{1}$$

where $\mathbf{z}_f$ represents spurious features that are easily fit but unstable (e.g., background noise correlated with labels in the SD) and $\mathbf{z}_s$ denotes specific style variations. $\boldsymbol{\epsilon}_f$, $\boldsymbol{\epsilon}_s$, and ϵ_y are mutually independent exogenous noise variables. Without explicit causal constraints, models trained via empirical risk minimization (ERM) tend to rely on $\mathbf{z}_f$ and $\mathbf{z}_s$ as shortcuts, leading to performance degradation on unseen TDs. Our objective is to recover the invariant causal mechanism $g_y(\cdot)$ while discarding spurious associations.

B. Structural Causal Models

To explicitly represent the causal relationships among features and labels, we employ SCMs [36]. Let $\mathcal{S} = \{s_i\}_{i=1}^{D+1}$ denote the set of causal factors, which comprises the D feature elements and the class label (indexed as $D+1$). An SCM defines the joint distribution over these factors as

$$p_{\mathcal{V}}(\mathbf{s}) = \prod_i p_{\mathcal{V}}(s_i|\mathbf{Pa}_i) \tag{2}$$

where $\mathbf{Pa}_i$ represents the set of parent nodes directly influencing the factor s_i. The SCM $\mathcal{V}$ entails a DAG $G = (\mathcal{S}, E)$, where the edge set E contains directed edges (e.g., $s_j \to s_i$) representing direct causal effects.

To enable end-to-end learning, we assume that the causal mechanisms can be approximated by a function class suitable for gradient-based optimization. We adopt a reconstruction-based paradigm where each factor s_i is predicted from its parents. Let $\mathbf{A} \in \mathbb{R}^{(D+1)\times(D+1)}$ be the weighted adjacency matrix. The structural equation is defined as

$$\hat{s}_i = g_i(\mathbf{s}) := \begin{cases} \mathbf{A}_i\mathbf{s}, & s_i \in \text{Numerical} \\ \mathbf{W}_{proj}\left(\mathbf{A}_{i,1:D}^{\top} \odot \mathbf{s}_{1:D}\right), & s_i \in \text{Categorical} \end{cases} \tag{3}$$

where $\mathbf{A}_i$ is the ith row of $\mathbf{A}$, masking nonparent nodes. For the categorical label node (s_{D+1}), the function outputs the classification logits via a projection $\mathbf{W}_{proj}$.

C. Differentiable Acyclicity and Global Causal Connectivity

Learning the structure $\mathbf{A}$ requires ensuring that the graph is a valid DAG. We utilize a continuous characterization based on the matrix exponential of the Hadamard product $\mathbf{M} = \mathbf{A}\odot\mathbf{A}$. The matrix exponential is defined by the Taylor series

$$e^{\mathbf{M}} = \mathbf{I} + \mathbf{M} + \frac{1}{2!}\mathbf{M}^2 + \frac{1}{3!}\mathbf{M}^3 + \cdots \tag{4}$$

This formulation serves two critical roles in our framework. Primarily, it imposes the acyclicity constraint. Since the trace $\mathrm{Tr}(\mathbf{M}^k)$ quantifies the presence of cycles of length k, a graph is a DAG if and only if $\mathrm{Tr}(\mathbf{M}^k) = 0$ for all $k \geq 1$. Consequently, the constraint is strictly derived as

$$h(\mathbf{A}) = \mathrm{Tr}\left(e^{\mathbf{A}\odot\mathbf{A}}\right) - (D+1) = 0. \tag{5}$$

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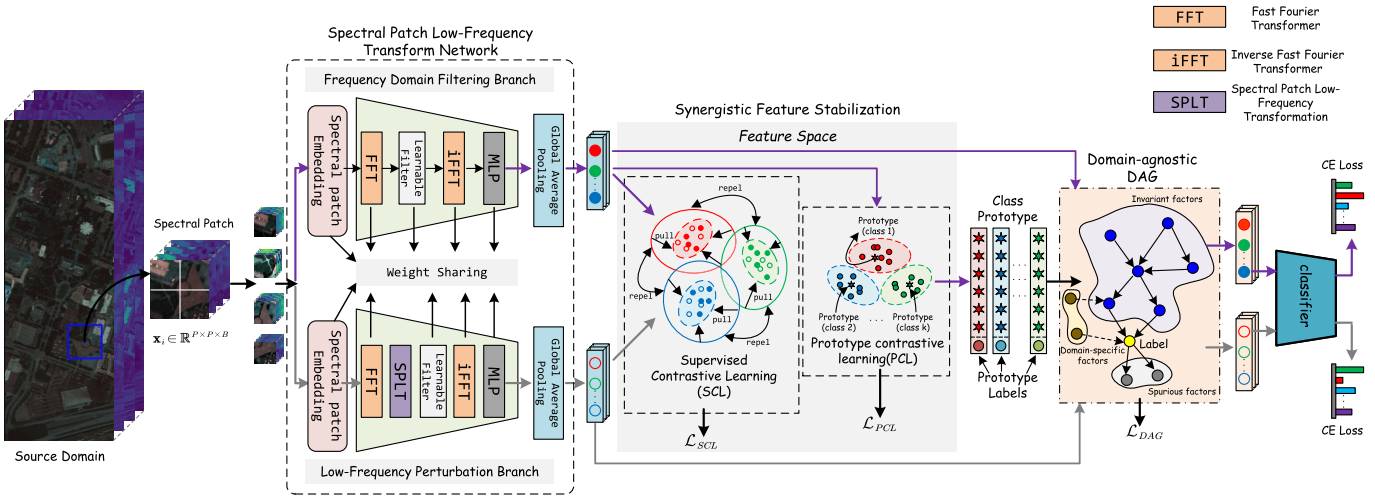


Fig. 1. Overview of the proposed Cross-SPECL framework. The workflow unfolds in a structured sequence: 1) dual-branch feature generation—an SPLT network constructs a main branch for feature extraction and an auxiliary branch with a nonlearnable SPLT module to generate perturbed features, simulating domain shifts; 2) SFS—SCL enforces invariance to these perturbations, while PCL aligns features to global centers, constructing highly representative class-conditional prototypes; and 3) domain-agnostic DAG learning and consistency: these stable prototypes drive the learning of a causal structure, generating a causal mask to filter out spurious correlations. This process is reinforced by a causal consistency constraint to ensure that the masked features are strictly domain-invariant. During inference, only the main branch and the learned causal mask are utilized for robust prediction.

Beyond ensuring structural validity, the matrix exponential simultaneously captures the global structural connectivity. The entry $(e^{\mathbf{M}})_{ij}$ aggregates the weights of all paths of arbitrary length from node j to node i, thereby reflecting the strength of potential causal pathways. By defining the global connectivity matrix as $\mathbf{E}_{total} = e^{\mathbf{A}\odot\mathbf{A}}$, we can quantify the aggregated causal influence. Unlike the adjacency matrix $\mathbf{A}$, which only models direct parents, $\mathbf{E}_{total}$ enables the identification of deep semantic dependencies by accounting for both direct and indirect causal connections, providing a robust global indicator for generating the causal mask.

IV. Proposed Method

For a single-source DG problem, the primary objective is to train a model exclusively on data from a single SD $\mathcal{H}_S$ such that it demonstrates robust performance on an unseen TD $\mathcal{H}_T$ during inference. The SD scene is formally defined as $\mathcal{H}_S = \{(\mathbf{x}_i, y_i)\}_{i=1}^{N_S}$, where $\mathbf{x}_i \in \mathbb{R}^{P\times P\times B}$ is a spectral patch and $y_i \in \mathcal{Y} \subset \mathbb{R}^K$ is the label.

Unlike prior single-source DG methods that rely on unstable adversarial training to diversify the SD distribution, our proposed Cross-SPECL constructs a cohesive, nonadversarial framework. It fundamentally shifts the paradigm from simple data augmentation to a structured process of "Stabilize-then-Disentangle." As illustrated in Fig. 1, Cross-SPECL orchestrates a synergistic interaction among feature extraction, contrastive alignment, and causal discovery. The core philosophy is to first establish a robust feature space resilient to spectral style variations via a dual-branch network and SFS. Building upon this stabilized foundation, the framework then explicitly mines the underlying causal structure to identify domain-invariant features while pruning domain-specific spurious correlations. This sequential integration ensures that the model learns generalized semantics rather than overfitting to source-specific biases.

A. SPLT Network

To effectively dissociate stable semantic information from domain-specific style variations, which predominantly manifest in the low-frequency components of spectral bands [37], we design the SPLT network.

1) Network Architecture and Sequential Dependencies: The proposed network adopts a dual-branch architecture. To clarify the data flow and variable dependencies, the processing sequence is defined as follows.

First, a raw spectral patch $\mathbf{x}_i \in \mathbb{R}^{P\times P\times B}$ sampled from the SD HSI is projected into a high-dimensional feature space via a shared *spectral patch embedding* module. This yields the intermediate embedding features, denoted as $\mathbf{h}_i \in \mathbb{R}^{H\times W\times D}$, where H, W, and D represent the height, width, and channel dimension of the features, respectively.

Subsequently, this common embedding $\mathbf{h}_i$ is fed into two parallel branches to generate the final feature representations $\mathbf{z}_i$.

1) *Frequency-Domain Filtering Branch (Main Branch):* This branch serves as the primary encoder. It takes $\mathbf{h}_i$ as input and processes it through frequency-domain layers to extract the native feature representation, denoted as $\mathbf{z}_i^{raw} \in \mathbb{R}^D$.
2) *LFP Branch (Auxiliary Branch):* This branch functions as a regularization auxiliary. It takes the same $\mathbf{h}_i$ as input but applies nonlearnable SPLT layers to the intermediate features. This generates an augmented feature representation, denoted as $\mathbf{z}_i^{aug} \in \mathbb{R}^D$.

Both branches share the same encoder parameters θ. The auxiliary branch incentivizes the main branch to disregard domain-specific noise during the subsequent contrastive learning phase by enforcing consistency between $\mathbf{z}_i^{raw}$ and $\mathbf{z}_i^{aug}$.

2) Frequency-Domain Processing: Both branches utilize a shared frequency-domain processing pipeline consisting of layer normalization (LN) [38], a fast Fourier transform (FFT)

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[32] layer, a learnable filter [39], an inverse FFT (iFFT) [40] layer, and an MLP.

The transformation of the intermediate embedding $\mathbf{h}_i$ within these blocks is detailed as follows. First, the FFT layer transforms $\mathbf{h}_i$ into the frequency domain

$$\mathcal{F}(\mathbf{h}_i)(m,n,d) = \sum_{h=0}^{H-1}\sum_{w=0}^{W-1} \mathbf{h}_i(h,w,d)\, e^{-2\pi j\left(\frac{h}{H}m+\frac{w}{W}n\right)} \tag{6}$$

where j is the imaginary unit ($j^2 = -1$). The indices m ($0 \le m < H$) and n ($0 \le n < W$) represent the discrete spatial frequency coordinates corresponding to the height H and width W, respectively, while d denotes the channel index. By default, the low-frequency components are shifted to the center of the frequency spectrum.

Subsequently, the frequency representations are modulated by a global learnable filter. This operation is expressed as

$$\tilde{\mathbf{h}}_i = \mathcal{F}(\mathbf{h}_i) \odot \mathbf{K} \tag{7}$$

where $\tilde{\mathbf{h}}_i$ denotes the filtered frequency-domain features, $\mathbf{K} \in \mathbb{R}^{H\times W\times D}$ represents the learnable weight matrix, and $\odot$ signifies the element-wise multiplication.

Finally, the filtered features $\tilde{\mathbf{h}}_i$ are converted back to the spatial domain via the iFFT layer, denoted as $\mathcal{F}^{-1}(\cdot)$, and processed by the MLP. The final output vectors $\mathbf{z}_i^{raw}$ (and similarly $\mathbf{z}_i^{aug}$) are obtained by applying global average pooling (GAP) and normalization to the processed feature maps.

3) Spectral Patch Low-Frequency Transformation: In the auxiliary branch, the SPLT layer simulates domain shifts by introducing controlled perturbations to the low-frequency components of the embedding features $\mathbf{h}_i$.

Specifically, for a batch of embedding features $\mathcal{H}_{bs} = [\mathbf{h}_1, \mathbf{h}_2, \ldots, \mathbf{h}_b]$, we first obtain the frequency-domain representation via FFT. Subsequently, the low-frequency components are isolated using a binary mask $\mathcal{M}$

$$\mathcal{F}_l(\mathcal{H}_{bs}) = \mathcal{F}(\mathcal{H}_{bs}) \odot \mathcal{M}. \tag{8}$$

Here, the binary mask $\mathcal{M}$ is defined such that the center region (low frequencies) is retained, while high frequencies are zeroed out

$$\mathcal{M}_{m,n} = \begin{cases} 1, & \text{if } \max\left(\left|m-\frac{H}{2}\right|, \left|n-\frac{W}{2}\right|\right) \leqslant \gamma\frac{\min(H,W)}{2} \\ 0, & \text{otherwise} \end{cases} \tag{9}$$

where $\gamma \in (0,1]$ is a hyperparameter determining the bandwidth of the low-frequency components.

We model the distribution of each element in the extracted low-frequency components $\mathcal{F}_l(\mathcal{H}_{bs})$ across the batch as a multivariate Gaussian distribution [33]. For a frequency component at index (m,n) in the dth channel, its mean is its original value, and its variance is calculated based on the batch statistics

$$\sigma^2(\mathcal{F}_l(\mathbf{h}_i)(m,n,d)) = \frac{1}{b}\sum_{k=1}^{b}\Big[\mathcal{F}_l(\mathbf{h}_k)(m,n,d) - \mathbb{E}_{\mathcal{F}_l(\mathcal{H}_{bs})}[\mathcal{F}_l(\mathbf{h}_k)(m,n,d)]\Big]^2 \tag{10}$$

where $\mathbb{E}_{\mathcal{F}_l(\mathcal{H}_{bs})}[\cdot]$ denotes the expectation (mean) across the batch.

To generate the augmented view, we resample the low-frequency value from this estimated distribution

$$\hat{\mathcal{F}}_l(\mathbf{h}_i) = \mathcal{F}_l(\mathbf{h}_i) + \beta \cdot \sigma(\mathcal{F}_l(\mathbf{h}_i)) \tag{11}$$

where β represents random noise sampled from a standard normal distribution $\mathcal{N}(0,1)$, possessing the same shape as the feature map $\mathcal{F}_l(\mathbf{h}_i)$ to enable element-wise broadcasting. Finally, the perturbed low-frequency components $\hat{\mathcal{F}}_l$ are combined with the original high-frequency components and transformed back to the spatial domain via iFFT. This process yields the perturbed intermediate features, which are subsequently processed to form the final augmented output $\mathbf{z}_i^{aug}$.

The rationale for this perturbation strategy is twofold, grounded in both the physical characteristics of hyperspectral domain shifts and frequency-domain theory. First, domain discrepancies in HSIs, typically induced by varying illumination conditions and imaging sensors, manifest as spatially smooth modulations that affect global intensity and spectral trends. In the frequency domain, these style-related attributes are predominantly encapsulated in the amplitude spectrum of the low-frequency components, as empirically demonstrated by the significant amplitude discrepancy shown in Fig. 2. Conversely, as visualized in Fig. 2(a), the high-frequency components encode fine-grained structural semantics, such as edges, texture patterns, and object boundaries, which define the semantic identity of the land cover class. Therefore, by exclusively perturbing the amplitude of the masked low-frequency region $\mathcal{F}_l$ while keeping the high-frequency components intact, we simulate spectral style shifts without corrupting the class semantics of the spectral patch. This compels the model to learn features that are invariant to specific style variations while preserving the essential semantic features for classification. Second, to ensure training stability, the perturbation magnitude is dynamically scaled by the batch-wise standard deviation $\sigma(\mathcal{F}_l)$. This distribution-aware mechanism constrains the augmented features within the statistical tolerance of the underlying feature manifold, effectively preventing the generated samples from drifting into invalid regions of the feature space.

B. Synergistic Feature Stabilization

Learning a DAG structure directly from individual samples across the entire dataset is computationally prohibitive and sensitive to sample-specific noise. To address this, we adopt a prototype-based strategy where the DAG structure is learned from class-conditional prototypes, which serve as a stable proxy for the SD distribution. Consequently, the reliability of the discovered causal structure heavily depends on the representativeness and robustness of these prototypes. To construct high-quality prototypes that are resilient to domain shifts, we propose a synergistic stabilization strategy combining SCL and PCL.

1) Supervised Contrastive Learning: First, we employ SCL to leverage the auxiliary branch for feature refinement. Since the auxiliary branch injects LFPs to simulate domain style variations, simply aggregating features to compute prototypes would incorporate this noise. SCL addresses this by

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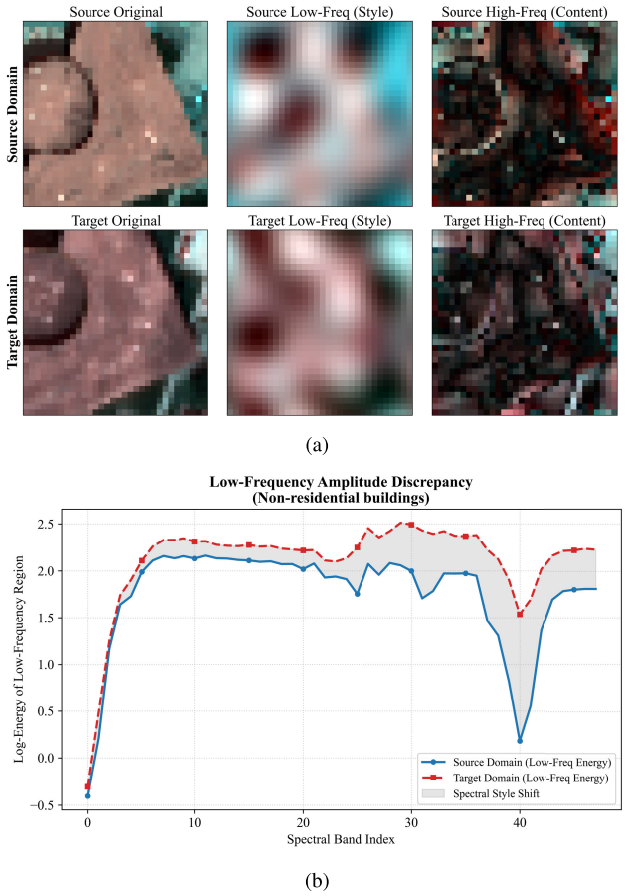


Fig. 2. Frequency-domain analysis of the "Nonresidential buildings" class (Houston 2013 versus 2018). (a) Spatial–spectral frequency decomposition: visual comparison shows domain shifts mainly affect low-frequency components (middle), while high-frequency components (right) preserve structural semantics. (b) Low-frequency amplitude discrepancy: quantitative analysis reveals significant spectral style shifts in low frequencies, motivating our SPLT strategy.

maximizing the similarity between the original features $\mathbf{z}_i^{raw}$ (main branch) and their perturbed counterparts $\mathbf{z}_i^{aug}$ (auxiliary branch) while clustering them with other same-class samples

$$\mathcal{L}_{scl} = \sum_{i \in I} \frac{-1}{|\mathcal{P}(i)|} \sum_{p \in \mathcal{P}(i)} \log \frac{\exp\left(\mathbf{z}_i \cdot \mathbf{z}_p / \tau\right)}{\sum_{a \in \mathcal{A}(i)} \exp\left(\mathbf{z}_i \cdot \mathbf{a} / \tau\right)}. \tag{12}$$

Minimizing this objective compels the frequency-domain filtering network to learn domain-invariant features that are robust to the low-frequency style variations introduced by the auxiliary branch. By effectively suppressing the intraclass variance caused by spectral shifts, SCL ensures that features form compact clusters, thereby providing a clean and stable foundation for computing class prototypes.

2) Prototype Contrastive Learning: Building upon the compact feature clusters facilitated by SCL, we employ PCL to further refine the representations. While SCL optimizes the relative distances between samples, it does not explicitly enforce alignment with a global reference. To bridge this gap, we enforce the raw features $\mathbf{z}^{raw}$ to align with their corresponding global class prototypes $\mathcal{P} = \{\mathbf{p}_k\}_{k=1}^K$

$$\mathcal{L}_{pcl} = -\frac{1}{b} \sum_{i=1}^{b} \log \frac{\exp\left(\mathbf{z}_i^{raw} \cdot \mathbf{p}_{y_i} / \tau\right)}{\sum_{k=1}^{K} \exp\left(\mathbf{z}_i^{raw} \cdot \mathbf{p}_k / \tau\right)}. \tag{13}$$

Simultaneously, to ensure temporal stability and smoothness, we employ a momentum update strategy for the prototypes. Specifically, the prototype for category y_i is updated as a moving average of the incoming features, followed by an L_2 normalization to project it onto the unit hypersphere

$$\mathbf{p}_{y_i} \leftarrow \text{Normalize}\left(m\mathbf{p}_{y_i} + (1-m)\,\mathbf{z}_i^{raw}\right) \tag{14}$$

where m is the momentum coefficient and Normalize$(\cdot)$ denotes L_2 normalization. This momentum-based mechanism ensures that the prototypes accumulate global distribution information over time, making them robust to batch-specific fluctuations. Consequently, these stabilized prototypes $\mathcal{P}$ serve as highly representative input nodes for the subsequent domain-agnostic DAG learning.

C. Domain-Agnostic DAG Learning

While the SFS strategy ensures that the learned prototypes $\mathcal{P} = \{\mathbf{p}_k\}_{k=1}^K$ are robust to low-frequency style variations, spurious correlations induced by high-frequency noise may still persist [22]. However, simply relying on statistical correlations is insufficient to distinguish between stable causal features and domain-specific spurious features. To address this, we propose to learn a domain-agnostic DAG structure directly from these stabilized prototypes. The rationale is twofold. First, prototypes naturally filter out sample-specific noise by aggregating class-wise commonalities. Second, and more importantly, by enforcing structural constraints (acyclicity and sparsity), the DAG optimization process is compelled to identify the most direct and parsimonious paths to the class label, thereby structurally discarding weak or indirect spurious associations.

1) SCM and Global Connectivity: We define the causal discovery problem over a set of variables $\mathcal{S}$ comprising the feature dimensions and the class label. Let $\mathbf{p}_k \in \mathbb{R}^D$ denote the stabilized feature vector of the kth class prototype. We construct a composite vector $\mathbf{u}_k = [\mathbf{p}_k, y_k] \in \mathbb{R}^{D+1}$, where the first D elements correspond to the features and the $(D+1)$th element corresponds to the label y_k. These variables are assumed to satisfy an SCM characterized by a weighted adjacency matrix $\mathbf{A} \in \mathbb{R}^{(D+1)\times(D+1)}$, where $[\mathbf{A}]_{i,j} \neq 0$ implies a direct causal influence $s_j \to s_i$.

Crucially, relying solely on the adjacency matrix $\mathbf{A}$ restricts the model to capturing only immediate, local relationships (direct parents). However, in a causal system, a variable s_i often influences the label y indirectly through intermediate variables. To capture this global influence flow, we leverage the global connectivity matrix $\mathbf{E}_{total} = e^{\mathbf{A}\odot\mathbf{A}}$, as defined in Section III-C. Specifically, the row segment corresponding to the label node, denoted as $\mathbf{e}_y = [\mathbf{E}_{total}]_{D+1,1:D}$, quantifies the cumulative causal contribution of each feature dimension to the target class y, aggregating paths of all lengths. This allows the model to identify deep semantic dependencies that might be missed by considering only direct connections.

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2) Causal Factor Reconstruction via Prototypes: To learn the adjacency matrix $\mathbf{A}$, we adopt a reconstruction-based paradigm grounded in the principle that if $\mathbf{A}$ accurately represents the underlying causal structure, each variable within the system should be reconstructible solely from its parents. Consequently, we aim to minimize a composite reconstruction objective $\mathcal{L}_{rec}$ that simultaneously enforces local structural consistency and global semantic predictiveness. At the feature level, we require that each dimension of the prototype vector be reconstructible from its direct parents as defined by the adjacency matrix, mathematically computed as $(\mathbf{A}\mathbf{u}_k)_{1:D}$. This constraint ensures that the learned graph captures the immediate dependencies among feature dimensions.

Complementing this local reconstruction, we leverage the global causal structure for the label prediction task. Unlike standard classification, which utilizes all features equally, we compute the predicted probability for the kth prototype by weighting the features according to their aggregated causal influence on the label node. This prediction mechanism is defined as

$$\hat{y}_k = \mathbf{W}_{rec}\left(\mathbf{e}_y^{\top} \odot \mathbf{p}_k\right) \tag{15}$$

where $\mathbf{e}_y^{\top}$ serves as a soft attention mask derived from the global causal graph and $\mathbf{W}_{rec}$ is a dedicated linear classifier. This formulation explicitly compels the model to base its classification decisions on features that possess significant causal pathways to the label, effectively filtering out spurious correlations. The overall reconstruction loss, integrating both the feature-level cosine similarity and the label-level prediction accuracy, is formulated as

$$\mathcal{L}_{rec} = \frac{1}{K}\sum_{k=1}^{K}\left(L_{\cos}\left((\mathbf{A}\mathbf{u}_k)_{1:D}, \mathbf{p}_k\right) + L_{ce}\left(\hat{y}_k, y_k\right)\right) \tag{16}$$

where $L_{\cos}$ denotes the cosine embedding loss measuring the fidelity of the structurally reconstructed features against the original prototype features $\mathbf{p}_k$ and L_{ce} represents the cross-entropy loss for the causal label prediction.

3) Optimization and Causal Masking: To ensure the learned graph is a valid DAG (i.e., no cycles), we enforce the exact acyclicity constraint based on the trace of the matrix exponential [41]: $\mathrm{Tr}(e^{\mathbf{A}\odot\mathbf{A}}) - (D+1) = 0$. The final learning objective for the domain-agnostic DAG is

$$\begin{aligned}\mathcal{L}_{dag} &= \mathcal{L}_{rec} + |\mathrm{vec}(\mathbf{A})|_1 \\ &\text{s.t. } \mathrm{Tr}\left(e^{\mathbf{A}\odot\mathbf{A}}\right) - (D+1) = 0\end{aligned} \tag{17}$$

where $|\cdot|_1$ imposes sparsity to prune redundant edges.

Once optimized, the matrix $\mathbf{A}$ effectively filters out spurious correlations. Since the sparsity constraint punishes redundant connections and spurious features typically exhibit weaker or less direct structural connections to the label compared to true semantic features, their corresponding weights in the DAG are suppressed. We extract the causal mask $\mathbf{m}_{dag}$ from the global connectivity matrix

$$\mathbf{m}_{dag} = [\mathbf{E}_{total}]_{D+1,1:D}^{\top}. \tag{18}$$

During inference, this mask is applied to the features $\mathbf{z}^{raw}$ via element-wise multiplication ($\mathbf{z}_{causal} = \mathbf{z}^{raw} \odot \mathbf{m}_{dag}$), effectively filtering out domain-specific spurious features and retaining only the causal factors for robust classification.

D. Causal Consistency and Optimization

Finally, to ensure that the model relies exclusively on the identified domain-invariant features for decision-making and to refine the geometric structure of the final feature space, we introduce a dual-consistency mechanism. This mechanism comprises prediction consistency for decision boundary alignment and CPA for feature space regularization.

1) Causal Prediction Consistency: Given the causal mask $\mathbf{m}_{dag}$ derived from the learned DAG structure [18], we apply this mask to the features from both the main branch ($\mathbf{z}_i^{raw}$) and the auxiliary branch ($\mathbf{z}_i^{aug}$). The underlying premise is that if $\mathbf{m}_{dag}$ accurately captures the true causal factors, it should effectively filter out both the naturally occurring domain-specific noise in the raw features and the artificially injected LFPs in the augmented features. Consequently, the classifier should produce consistent and accurate predictions for both masked representations, regardless of the spectral style variations. The prediction consistency loss is formally defined as

$$\begin{aligned}\mathcal{L}_{con} = \frac{1}{b}\sum_{i=1}^{b}\Big(&L_{ce}\left(\mathbf{W}_{cls}\left(\mathbf{z}_i^{raw} \odot \mathbf{m}_{dag}\right), y_i\right) \\ &+ L_{ce}\left(\mathbf{W}_{cls}\left(\mathbf{z}_i^{aug} \odot \mathbf{m}_{dag}\right), y_i\right)\Big)\end{aligned} \tag{19}$$

where $\mathbf{W}_{cls}$ represents the shared main linear classifier used for final inference. This objective serves to train the classifier to utilize only the causal features for decision-making while simultaneously verifying the validity of the causal mask.

2) Causal Prototype Alignment: While prediction consistency optimizes the decision boundary, it does not explicitly enforce the geometric compactness of the discovered causal features. To address this, we establish a second set of prototypes, termed causal prototypes $\mathcal{P}_{causal} = \{\mathbf{p}_k^{causal}\}_{k=1}^{K}$. These prototypes are dynamically updated via a momentum strategy using the masked raw features ($\mathbf{z}^{raw} \odot \mathbf{m}_{dag}$), thereby capturing the idealized distribution of the domain-invariant semantics. To ensure that the extracted causal features form compact, well-separated clusters, we apply a causal prototype contrastive loss (Causal PCL) directly to the masked raw features

$$\mathcal{L}_{pcl}^{causal} = -\frac{1}{b}\sum_{i=1}^{b}\log\frac{\exp\left(\left(\mathbf{z}_i^{raw} \odot \mathbf{m}_{dag}\right)\cdot \mathbf{p}_{y_i}^{causal}/\tau\right)}{\sum_{k=1}^{K}\exp\left(\left(\mathbf{z}_i^{raw} \odot \mathbf{m}_{dag}\right)\cdot \mathbf{p}_k^{causal}/\tau\right)}. \tag{20}$$

By minimizing this objective, the model forces the final masked representation to align closely with the stable causal prototypes. This process effectively regularizes the feature space, ensuring that the identified causal features are not only predictive but also possess high intraclass compactness and interclass separability, which is crucial for robust generalization.

3) Overall Optimization: The total objective function of Cross-SPECL integrates the progressive contrastive feature stabilization, DAG structure learning, and the final causal consistency mechanisms

$$\min_{\mathbf{A},\theta}\mathcal{L}_{total} = \mathcal{L}_{con} + \lambda_{DAG}\mathcal{L}_{dag} + \lambda_{CL}\left(\mathcal{L}_{scl} + \mathcal{L}_{pcl} + \mathcal{L}_{pcl}^{causal}\right). \tag{21}$$

Here, θ encompasses the parameters of the encoder (shared by both branches), the learnable weights of the DAG adjacency matrix $\mathbf{A}$, and the parameters of both classifiers

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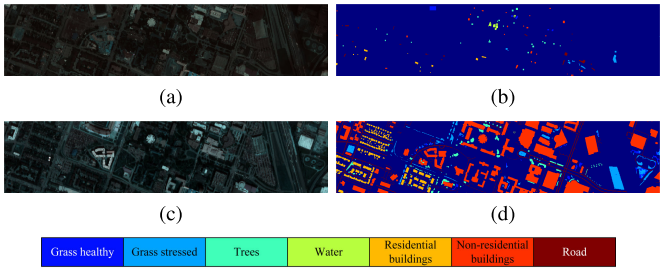


Fig. 3. Pseudo-color images and ground truth for the Houston 2013 and Houston 2018 datasets. (a) Pseudo-color image of Houston 2013, (b) ground truth of Houston 2013, (c) pseudo-color image of Houston 2018, and (d) ground truth of Houston 2018.

TABLE I
CATEGORIES AND NUMBER OF SAMPLES IN THE HOUSTON 2013 AND HOUSTON 2018 DATASETS

Class		Number of Samples	
No.	Name	Houston 2013 (Source)	Houston 2018 (Target)
1	Grass healthy	345	1353
2	Grass stressed	365	4888
3	Trees	365	2766
4	Water	285	22
5	Residential buildings	319	5347
6	Non-residential buildings	408	32459
7	Road	443	6365
Total		2530	53200

$\{\mathbf{W}_{cls}$ and $\mathbf{W}_{rec}\}$. The hyperparameters λ_{DAG} and λ_{CL} balance the contributions of the structure learning and the contrastive feature stabilization, respectively. This unified optimization strategy systematically guides the model from feature stabilization ("Stabilize") to causal structure discovery and consistent prediction ("Disentangle"), ensuring robust generalization on unseen domains.

V. EXPERIMENT RESULTS AND ANALYSIS

A. Datasets Description

1) Houston 2013 and Houston 2018 Datasets: The Houston 2013 [42] (144 bands, 349 $\times$ 1905 pixels, and 2.5-m resolution) and Houston 2018 [43] (48 bands, 1202 $\times$ 4172 pixels, and 1-m resolution) datasets cover the University of Houston. Both span 380–1050 nm. For experiments, we use an overlapping region of 209 $\times$ 955 pixels. The Houston 2018 data are downsampled to 2.5-m resolution, and 48 corresponding bands from Houston 2013 are selected, matching Houston 2018 [44]. Houston 2013 serves as the SD and Houston 2018 as the TD. Both datasets share seven identical land cover categories, as detailed in Table I and visualized in Fig. 3.

2) Pavia University and Pavia Center Datasets: These datasets were collected by the ROSIS sensor over northern Italy. The Pavia University dataset (610 $\times$ 340 pixels, 103 bands) and the Pavia Center dataset (1096 $\times$ 715 pixels, 102 bands) both exhibit a 1.3-m spatial resolution and cover the spectral range of 430–860 nm. The last band of Pavia University is removed to match Pavia Center's 102 bands. Pavia University is utilized as the SD and Pavia Center as the TD. Both datasets encompass seven common land cover classes, as presented in Table II and Fig. 4.

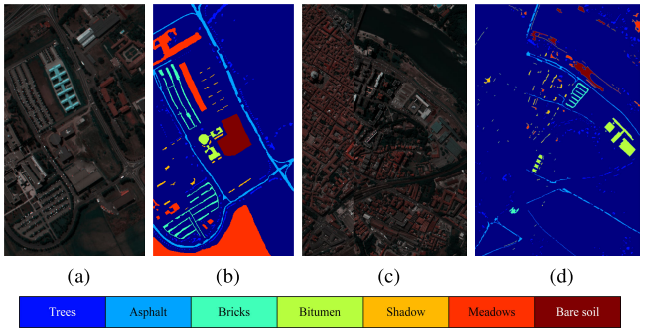


Fig. 4. Pseudo-color images and ground truth for the Pavia University and Pavia Center datasets. (a) Pseudo-color image of Pavia University, (b) ground truth of Pavia University, (c) pseudo-color image of Pavia Center, and (d) ground truth of Pavia Center.

TABLE II
CATEGORIES AND NUMBER OF SAMPLES IN THE PAVIA UNIVERSITY AND PAVIA CENTER DATASETS

Class		Number of Samples	
No.	Name	Pavia University (Source)	Pavia Center (Target)
1	Tree	3064	7598
2	Asphalt	6631	9248
3	Brick	3682	2685
4	Bitumen	1330	7287
5	Shadow	947	2863
6	Meadows	18649	3090
7	Bare soil	5029	6584
Total		39332	39355

TABLE III
CLASSES AND SAMPLE NUMBERS IN THE HANGZHOU AND SHANGHAI DATASETS

Class		Number of Samples	
No.	Name	Hangzhou (Source)	Shanghai (Target)
1	Water	18043	123123
2	Land/Building	77450	161689
3	Plant	40207	83188
Total		135700	368000

3) Hangzhou and Shanghai Datasets: The Hangzhou and Shanghai datasets were acquired by the EO-1 Hyperion hyperspectral sensor over the Hangzhou and Shanghai regions in China, respectively. These datasets possess 198 spectral bands, with spatial dimensions of 590 $\times$ 230 and 1600 $\times$ 230 pixels. The shared water, land/building, and vegetation categories in both scenes are utilized as classification classes, and the number of samples for each category is listed in Table III. In the experiments, the Hangzhou data serve as the SD, and the Shanghai data serve as the TD. The pseudo-color images and the corresponding ground truth maps of these two scenes are shown in Fig. 5.

B. Experimental Setup

1) Comparison Methods: To evaluate the single-source DG performance of Cross-SPECL, seven state-of-the-art transfer learning methods were used for cross-scene HSI classification performance comparison. These include two classic DA methods: deep subdomain adaptation network (DSAN) [45] and TSTnet [16]. Additionally, five state-of-the-art single-source

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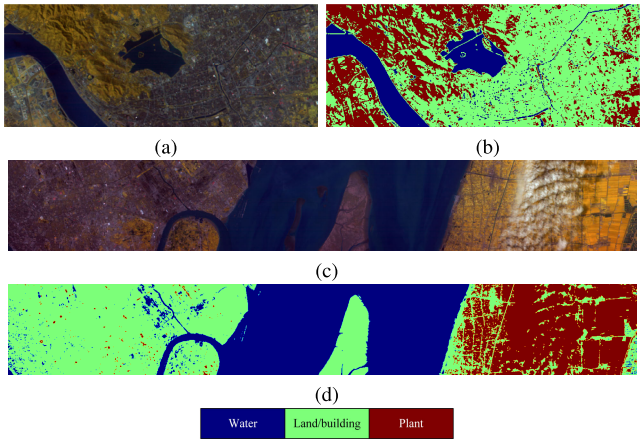


Fig. 5. Pseudo-color images and ground truth for the Hangzhou and Shanghai datasets. (a) Pseudo-color image of Hangzhou. (b) Ground truth of Hangzhou. (c) Pseudo-color image of Shanghai. (d) Ground truth of Shanghai.

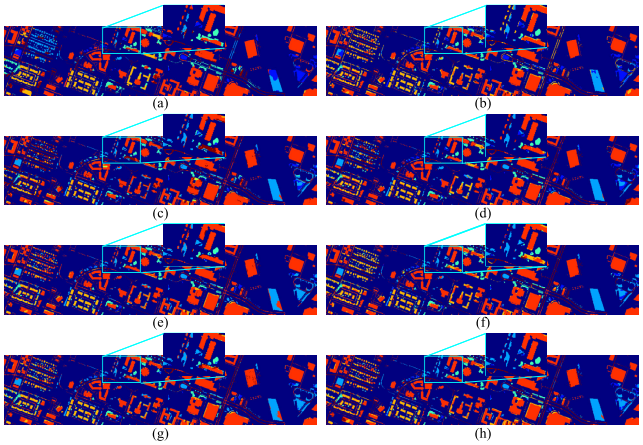


Fig. 6. Classification result maps of Cross-SPECL and the seven state-of-the-art comparison methods on the Houston 2018 dataset. (a) DSAN, (b) TSTnet, (c) PDEN, (d) LDGnet, (e) SDEnet, (f) S^{2}AMSnet, (g) FDGNet, and (h) Cross-SPECL.

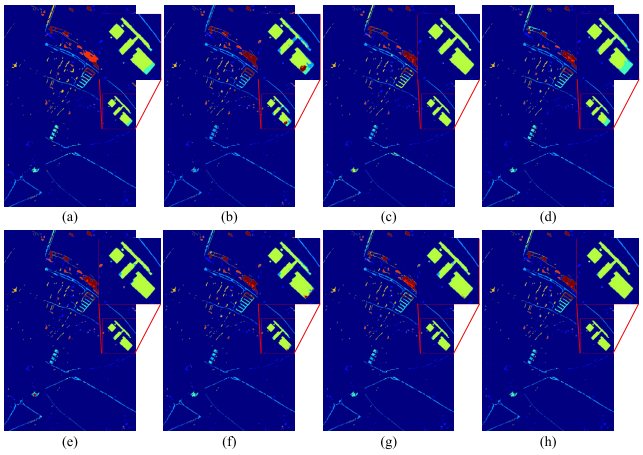


Fig. 7. Classification result maps of Cross-SPECL and the seven state-of-the-art comparison methods on the Pavia Center dataset. (a) DSAN, (b) TSTnet, (c) PDEN, (d) LDGnet, (e) SDEnet, (f) S^{2}AMSnet, (g) FDGNet, and (h) Cross-SPECL.

DG methods were employed: progressive domain expansion network (PDEN) [46], language-aware DG network (LDGnet) [47], SDEnet [17], spectral–spatial adversarial multidomain synthesis network (S^{2}AMSnet) [20], and frequency disentanglement and data geometry network (FDGNet) [48].

2) Evaluation Criteria: The cross-scene HSI classification performance was evaluated using both qualitative and quantitative measures. The qualitative metric involved visual assessment of classification result maps obtained from the target scenes. Quantitative metrics included class accuracy (CA), overall accuracy (OA), average accuracy (AA), and the Kappa coefficient (κ).

3) Experimental Details: The hyperparameter configurations for the comparative methods are standardized to ensure a fair evaluation. Specifically, DSAN employs spectral patches of size 32×32 to align with the input specifications of the ResNet18 feature extraction backbone. In contrast, PDEN, LDGnet, SDEnet, S^{2}AMSnet, and FDGNet achieve optimal performance utilizing a spectral patch dimension of 13×13. All other internal parameters for these baseline methods adhere strictly to the optimal settings reported in their original literature.

For the proposed Cross-SPECL framework, the optimal parameter settings vary slightly across the three experimental scenarios, namely the transfer tasks from Houston 2013 to Houston 2018, Pavia University to Pavia Center, and Hangzhou to Shanghai. The input spectral patch size is consistently set to 8×8 for all three tasks. The learning rates are initialized at 2.5×10^{-4}, 2.5×10^{-4}, and 1×10^{-3}. Within the SFS mechanism, the scalar temperature parameters for both the SCL and PCL losses are configured as 0.07, 0.07, and 0.1 for the respective tasks. The momentum parameter m, which governs the update rate of class-conditional prototypes, is fixed at 0.99 across all experiments. Furthermore, as the batch size determines the quantity of negative samples in SCL and influences the domain-invariant feature extraction capability of the network, it is set to 256 for the Houston task and 4096 for both the Pavia and Hangzhou to Shanghai tasks. Finally, the hyperparameters λ_{DAG} and λ_{CL}, designed to balance the DAG reconstruction loss and the contrastive learning objectives, are assigned values of 0.1 and 0.1, 0.1 and 0.5, and 0.1 and 0.1 for the three corresponding transfer scenarios.

4) Experimental Environment: All experiments were conducted on a system equipped with an Intel Core i9-14900k 24-core CPU processor and an NVIDIA GeForce RTX 4090 GPU with 24 GB of memory. The proposed Cross-SPECL and all comparison methods were implemented using Python 3.8.3 and PyTorch 1.11.

C. Performance on Cross-Scene HSI Classification

We first qualitatively evaluate the classification results of Cross-SPECL and seven state-of-the-art comparison methods on the Houston 2018, Pavia Center, and Shanghai target scenes. Figs. 6–8 display the classification maps generated by the different methods. In these visualization results, each labeled pixel is mapped to its predicted class, while unlabeled pixels are rendered as background. Compared to the ground truth, the maps generated by Cross-SPECL exhibit significantly reduced noise and smoother homogeneous regions

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TABLE IV

CLASSIFICATION RESULTS OF CROSS-SPECL AND THE COMPARATIVE METHODS ON THE HOUSTON 2018 DATASET

Class	Classification Methods							
	DSAN [45]	TSTnet [16]	PDEN [46]	LDGnet [47]	SDEnet [17]	S^{2}AMSnet [20]	FDGNet [48]	Cross-SPECL
1	61.86	93.94	54.62	77.90	32.96	49.52	36.73	64.82
2	27.21	52.95	79.11	77.82	78.87	85.29	83.78	91.43
3	57.01	49.13	51.95	65.04	61.28	63.52	61.10	67.93
4	81.82	100	100	81.82	100	100	100	100
5	55.10	80.36	46.29	59.34	63.64	84.05	69.63	68.36
6	87.33	88.31	86.74	90.65	90.92	89.82	90.47	89.20
7	43.41	40.39	63.74	54.78	49.13	51.15	50.64	60.09
OA (%)	71.08	76.64	76.60	80.38	79.06	81.81	80.10	**82.11**
AA (%)	59.11	72.15	68.92	72.48	68.11	74.76	70.34	**77.40**
$\kappa \times 100$	51.74	61.45	60.96	66.30	63.11	69.34	65.71	**69.61**

TABLE V

CLASSIFICATION RESULTS OF CROSS-SPECL AND THE COMPARATIVE METHODS ON THE PAVIA CENTER DATASET

Class	Classification Methods							
	DSAN [45]	TSTnet [16]	PDEN [46]	LDGnet [47]	SDEnet [17]	S^{2}AMSnet [20]	FDGNet [48]	Cross-SPECL
1	93.43	81.94	94.16	97.79	82.48	87.01	90.80	93.59
2	81.35	96.30	83.09	87.11	82.42	92.40	85.22	87.37
3	66.78	20.19	19.59	64.62	51.25	69.31	65.77	70.91
4	78.84	65.73	84.85	70.34	85.04	79.98	85.38	86.59
5	89.21	91.41	80.20	90.22	80.54	89.87	85.85	89.03
6	87.12	86.63	63.56	56.50	82.75	77.48	70.81	73.50
7	56.24	91.65	91.92	78.54	91.65	91.10	79.74	92.44
OA (%)	79.05	80.78	80.95	80.92	82.22	85.91	83.00	**87.18**
AA (%)	79.00	76.26	73.91	77.87	79.45	83.88	80.51	**84.78**
$\kappa \times 100$	75.20	76.72	76.84	77.06	78.62	83.02	79.54	**84.55**

TABLE VI

CLASSIFICATION RESULTS OF CROSS-SPECL AND THE COMPARATIVE METHODS ON THE SHANGHAI DATASET

Class	Classification Methods							
	DSAN [45]	TSTnet [16]	PDEN [46]	LDGnet [47]	SDEnet [17]	S^{2}AMSnet [20]	FDGNet [48]	Cross-SPECL
1	90.16	89.31	94.87	94.47	93.87	90.16	95.62	95.58
2	77.53	81.74	58.47	48.73	65.73	86.44	81.71	90.26
3	99.50	99.28	99.99	99.97	99.95	99.12	99.28	94.36
OA (%)	86.72	88.24	80.03	75.62	82.88	90.55	90.34	**92.97**
AA (%)	89.06	86.78	84.44	81.06	86.52	91.91	92.20	**93.40**
$\kappa \times 100$	79.94	82.10	70.68	64.66	74.59	85.53	85.34	**89.15**

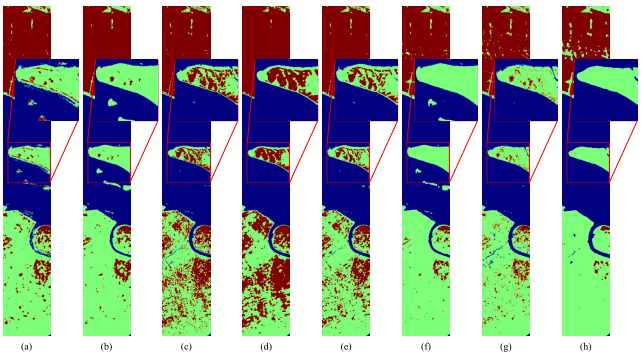


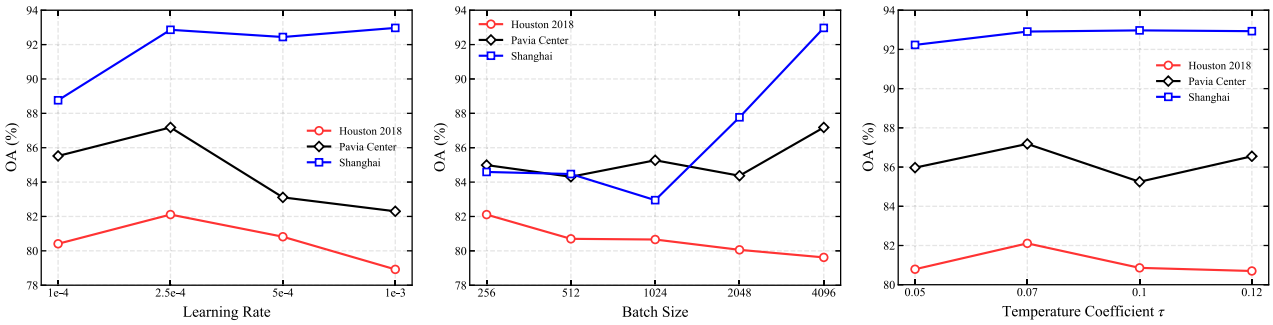
Fig. 8. Classification result maps of Cross-SPECL and the seven state-of-the-art comparison methods on the Shanghai dataset. (a) DSAN, (b) TSTnet, (c) PDEN, (d) LDGnet, (e) SDEnet, (f) S^{2}AMSnet, (g) FDGNet, and (h) Cross-SPECL.

across all three target scenes. Specifically, Cross-SPECL produces more coherent predictions for challenging classes that are often misclassified by comparison methods due to severe spectral shifts. Notable examples include *Grass Stressed* (Class 2) in Houston 2018, *Brick* (Class 3) and *Bare Soil* (Class 7) in Pavia Center, and *Land/Building* (Class 2) in the Shanghai dataset. This visual superiority validates that our method effectively preserves semantic structure even in the presence of significant domain discrepancies.

Next, to quantitatively assess the cross-scene classification performance, we conducted a comprehensive evaluation on the Houston 2018, Pavia Center, and Shanghai datasets. Tables IV–VI summarize the quantitative results, including CA, OA, AA, and the κ.

Analyzing the baseline methods reveals distinct performance patterns. Among the DA methods, TSTnet consistently outperforms DSAN across all three datasets. This suggests that TSTnet, which is specifically tailored for HSI characteristics, effectively leverages spectral information to align distributions, whereas DSAN, originally designed for RGB images, struggles to handle the high-dimensional spectral shifts inherent in HSIs. Among the single-source DG methods, S^{2}AMSnet generally achieves the second-best performance. Notably, S^{2}AMSnet surpasses the best-performing DA method (TSTnet) by margins of 5.17%, 5.13%, and 2.31% in terms of OA on the Houston 2018, Pavia Center, and Shanghai

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(a) (b) (c)

Fig. 9. Parameter sensitivity analysis. (a) Learning rate, (b) batch size, and (c) temperature parameter.

datasets, respectively. This finding reinforces the premise that robust DG models can learn transferable representations solely from the SD, eliminating the dependency on TD data during training.

Most importantly, the proposed Cross-SPECL consistently achieves state-of-the-art performance across all three cross-scene tasks, demonstrating exceptional generalization capabilities. Specifically, compared to the best-performing DA method (TSTnet), Cross-SPECL yields substantial OA improvements of 5.47%, 6.40%, and 4.73% on the Houston 2018, Pavia Center, and Shanghai datasets, respectively. Furthermore, compared to the strongest DG competitor, S^2AMSnet, Cross-SPECL achieves OA gains of 0.30%, 1.27%, and 2.42%. While methods like SDEnet and S^2AMSnet attempt to improve robustness by diversifying the source distribution via adversarial training or generation, they risk overfitting to expanded noise or spurious correlations. In contrast, the superior performance of Cross-SPECL can be attributed to its unique "Stabilize-then-Disentangle" paradigm. By utilizing the SPLT network to filter out domain-specific style variations and employing the Domain-Agnostic DAG to explicitly prune spurious causal features, Cross-SPECL extracts strictly domain-invariant semantics. Moreover, by avoiding unstable adversarial training, Cross-SPECL ensures a more reliable and consistent optimization process, leading to robust adaptation to unseen scenes.

D. Parameter Analysis

In this section, we conduct a systematic sensitivity analysis of the key hyperparameters in Cross-SPECL to evaluate their impact on model performance across the three cross-scene classification tasks. The hyperparameters investigated include the input spectral patch size, learning rate, batch size, temperature parameter τ, and the loss balancing weights λ_{DAG} and λ_{CL}.

1) Impact of Patch Size: The patch size determines the spatial receptive field and the amount of contextual information available to the model. We evaluated patch sizes from the set $\{8 \times 8, 16 \times 16, 32 \times 32\}$. As summarized in Table VII, Cross-SPECL consistently achieves optimal performance with a patch size of 8×8 across all three datasets (Houston 2018, Pavia Center, and Shanghai). While larger patches provide broader context, they may also introduce irrelevant background noise that interferes with the extraction of fine-grained, domain-invariant features. The results suggest that an 8×8 patch size offers the best tradeoff, capturing sufficient local spatial semantics while maintaining the purity of the spectral signatures required for accurate causal discovery.

TABLE VII
IMPACT OF PATCH SIZE ON CROSS-SPECL'S CROSS-SCENE HSI CLASSIFICATION RESULTS

Patch Size	8	16	32
Houston 2018	**82.11**	76.13	76.68
Pavia Center	**87.18**	76.72	67.07
Shanghai	**92.97**	90.27	88.45

2) Impact of Learning Rate: The learning rate controls the step size during optimization. We explored learning rates in the range of $[1e^{-4}, 1e^{-3}]$. As illustrated in Fig. 9(a), the optimal learning rate varies slightly by dataset. For the Houston 2018 and Pavia Center tasks, the model achieves peak OA at 2.5×10^{-4}. Conversely, for the Shanghai task, a larger learning rate of 1×10^{-3} yields the best performance (92.97%), suggesting that this dataset may require larger update steps to escape local minima or converge efficiently.

3) Impact of Batch Size: Batch size is a critical hyperparameter for contrastive learning frameworks (SCL and PCL), as it determines the number of negative samples available for optimization. We tested batch sizes ranging from 256 to 4096. Fig. 9(b) reveals distinct pReferences across datasets. For the Houston 2018 dataset, a smaller batch size of 256 is sufficient to achieve optimal results (82.11%). In contrast, both the Pavia Center and Shanghai datasets benefit significantly from larger batch sizes, achieving their highest OAs of 87.18% and 92.97%, respectively, at a batch size of 4096. This indicates that for datasets with larger sample volumes or more complex distributions, increasing the pool of negative samples enhances the quality of the learned representations.

4) Impact of Temperature Parameter \tau: The temperature parameter τ modulates the sharpness of the probability distribution in contrastive losses. We varied τ within $[0.05, 0.12]$. As shown in Fig. 9(c), the model is relatively sensitive to this parameter. For Houston 2018 and Pavia Center, a smaller temperature of $\tau = 0.07$ proves optimal, facilitating a sharper discrimination between positive and negative pairs. For the Shanghai dataset, a slightly smoother distribution with $\tau = 0.1$ results in the best performance (92.97%).

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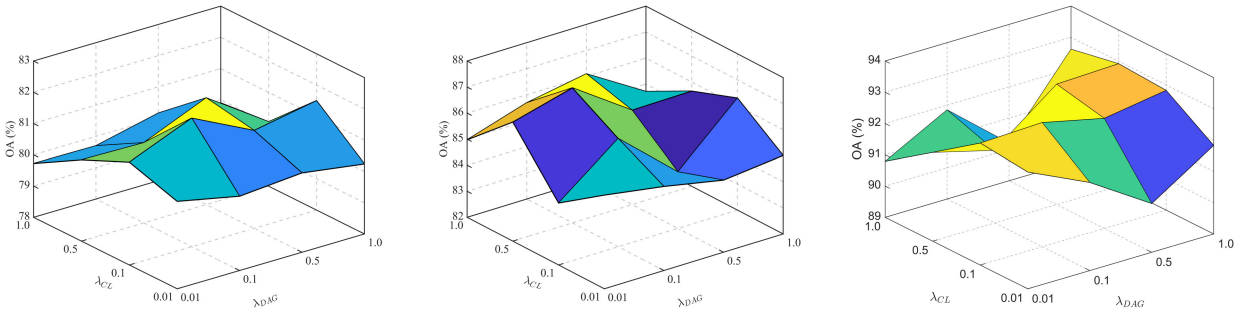


Fig. 10. Impact of λ_{DAG} and λ_{CL} on the cross-scene HSI classification performance of Cross-SPECL. (a) Houston 2018, (b) Pavia Center, and (c) Shanghai.

TABLE VIII
IMPACT OF BANDWIDTH RATIO γ ON CROSS-SCENE CLASSIFICATION ACCURACY (OA %)

Bandwidth Ratio γ	**0**	**0.5**	**1.0**
Houston 2018	79.33	**82.11**	80.71
Pavia Center	84.82	**87.18**	84.99
Shanghai	88.88	**92.97**	90.66

5) Sensitivity to Loss Components (λ_{DAG} and λ_{CL}): Finally, we analyzed the tradeoff between the DAG reconstruction loss and the progressive contrastive learning terms, controlled by λ_{DAG} and λ_{CL}, respectively. We performed a grid search with values in $\{0.01, 0.1, 0.5, 1.0\}$. Fig. 10 depicts the joint impact of these weights. For the Houston 2018 task, the best balance is achieved with $\lambda_{DAG} = 0.1$ and $\lambda_{CL} = 0.1$. For the Pavia Center task, a higher weight on contrastive stabilization proves beneficial, with the optimal setting being $\lambda_{DAG} = 0.1$ and $\lambda_{CL} = 0.5$. For the Shanghai task, the model performs best with equal weights of $\lambda_{DAG} = 0.1$ and $\lambda_{CL} = 0.1$. These results indicate that while both components are essential, the optimal balance depends on the specific characteristics of the SD and TD shift.

6) Impact of the Bandwidth Ratio \gamma: We investigate the impact of the bandwidth ratio γ, which controls the scope of spectral perturbation, by evaluating three topological settings: $\gamma = 0$ (no perturbation), $\gamma = 0.5$ (LFP), and $\gamma = 1.0$ (global perturbation). As summarized in Table VIII, the baseline configuration with $\gamma = 0$ yields suboptimal results, achieving an OA of only 79.33% on the Houston 2018 dataset. In contrast, the model consistently achieves optimal performance at $\gamma = 0.5$, where the OA increases significantly to 82.11%. This improvement stems from our *distribution-aware perturbation* strategy, formally defined as $\hat{\mathcal{F}}_l = \mathcal{F}_l + \beta \cdot \sigma(\mathcal{F}_l)$. By dynamically scaling the random noise $\beta \sim \mathcal{N}(0, 1)$ with the batch-wise standard deviation $\sigma(\mathcal{F}_l)$, we ensure that augmented features remain within the statistical tolerance of their respective classes. This mechanism effectively simulates realistic spectral shifts without causing feature manifold drift. However, extending perturbation to the full spectrum ($\gamma = 1.0$) leads to a decline in performance, reducing the OA to 80.71%. This degradation occurs because global perturbation inevitably corrupts high-frequency components that encode essential spatial semantics like edges. Thus, $\gamma = 0.5$ strikes the necessary balance between enforcing style invariance and preserving semantic integrity.

TABLE IX
ABLATION STUDY OF DIFFERENT COMPONENTS IN THE PROPOSED CROSS-SPECL FRAMEWORK

Configuration	**Loss Functions**				**OA (%)**	
	$\mathcal{L}_{scl}$	$\mathcal{L}_{dag}$	$\mathcal{L}_{pcl}$	$\mathcal{L}_{pcl}^{causal}$	Houston 2018	Pavia Center
Source Only	✗	✗	✗	✗	75.83	79.72
w/o LFP branch	✗	✓	✓	✓	79.33	84.82
w/o DAG	✓	✗	✗	✗	80.05	84.93
w/o SFS	✗	✓	✗	✓	79.42	85.50
w/o CPA	✓	✓	✓	✗	81.21	85.65
Cross-SPECL	✓	✓	✓	✓	**82.11**	**87.18**

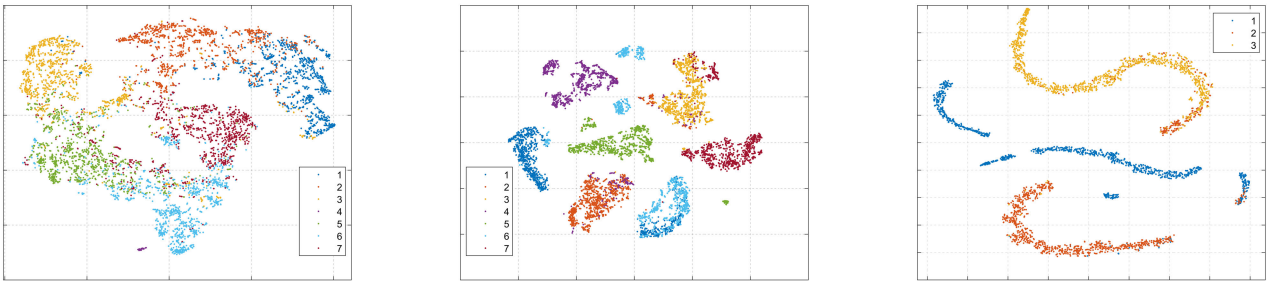
Note: LFP: Low-Frequency Perturbation; DAG: Directed Acyclic Graph; SFS: Synergistic Feature Stabilization; CPA: Causal Prototype Alignment.

E. Ablation Studies

To systematically validate the effectiveness of the proposed Cross-SPECL framework and quantify the individual contribution of each key component, we conducted comprehensive ablation studies on the Houston 2018 and Pavia Center datasets. The experimental results, derived by systematically removing specific modules from the full objective function, are summarized in Table IX. The baseline "Source Only" model, which is trained exclusively on SD data via ERM, yields suboptimal performance. This highlights the inherent difficulty of generalizing to unseen domains without explicit constraints. In contrast, the complete Cross-SPECL framework achieves the highest accuracy across all tasks, demonstrating the necessity of the proposed synergistic design.

We first investigate the impact of the feature stabilization components. The LFP branch is pivotal for simulating domain shifts; as evidenced in the second row of Table IX, its removal ("w/o LFP branch") leads to a performance degradation of 2.78% and 2.36% in OA on the Houston 2018 and Pavia Center datasets, respectively. This indicates that without the auxiliary perturbations, the model struggles to learn representations robust to spectral style variations. Similarly, the SFS mechanism, which integrates SCL and PCL, is crucial for constructing representative prototypes. Removing this component

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(a) (b) (c)

Fig. 11. Class separability proposed by Cross-SPECL. (a) Feature distribution of Houston 2018 extracted by Cross-SPECL. (b) Feature distribution of Pavia Center extracted by Cross-SPECL. (c) Feature distribution of Shanghai extracted by Cross-SPECL.

TABLE X
TOTAL NUMBER OF TRAINABLE PARAMETERS AND EXECUTION TIME PER EPOCH FOR DIFFERENT METHODS

Dataset	Metric	*Cross-Scene Hyperspectral Image Classification Methods*							
		DSAN [45]	**TSTnet** [16]	**PDEN** [46]	**LDGnet** [47]	**SDEnet** [17]	**S^{2}AMSnet** [20]	**FDGNet** [48]	**Cross-SPECL**
Houston 2013	Params (M)	16.54	7.78	1.55	35.88	1.89	1.07	3.29	**1.03**
	Time (s)	18.78	9.71	6.55	20.18	5.16	10.57	5.97	**1.10**
Pavia University	Params (M)	16.57	7.80	1.66	36.81	2.53	**0.94**	4.23	1.14
	Time (s)	87.96	35.01	23.26	71.67	18.92	29.23	26.45	**13.69**
Hangzhou	Params (M)	16.34	7.81	1.85	38.38	3.68	1.44	6.40	**1.34**
	Time (s)	701.93	158.69	103.08	308.23	82.69	83.69	216.97	**59.74**

("w/o SFS") results in a significant drop in OA to 79.42% on the Houston 2018 dataset. This decline suggests that without the dual-contrastive constraints, the learned prototypes are susceptible to noise, thereby failing to provide a stable foundation for the subsequent causal structure discovery.

Furthermore, we analyze the contribution of the causal disentanglement modules. The domain-agnostic DAG learning module is designed to identify and prune spurious correlations. As shown in the third row of Table IX ("w/o DAG"), omitting this module causes the OA to decrease by 2.39% and 2.25% on the Houston 2018 and Pavia Center datasets, respectively. This confirms that simply learning stable features is insufficient; explicitly modeling the causal structure to generate a causal mask is essential for filtering out nontransferable features. Finally, the CPA serves to refine the geometric structure of the masked features. The exclusion of this term ("w/o CPA") leads to a performance dip of 1.45% on the Houston 2018 dataset, verifying that enforcing intraclass compactness on the causal features further enhances the model's discriminability on unseen domains. Collectively, these results unequivocally demonstrate that the seamless integration of feature stabilization and causal disentanglement is indispensable for achieving robust single-source DG.

F. Generalization Performance of Cross-SPECL

To verify whether Cross-SPECL, trained exclusively on a single SD, has successfully learned domain-invariant representations that generalize to unseen TDs, we utilize t-distributed stochastic neighbor embedding (t-SNE) to visualize the feature distributions on the target scenes. The core premise is that a robust DG model should map unseen target data into a feature space where samples from the same class are tightly clustered (high intraclass compactness) and samples from different classes are clearly separated (high interclass separability), despite the presence of domain shifts.

Fig. 11 presents the t-SNE visualizations of the features extracted by Cross-SPECL on the Houston 2018, Pavia Center, and Shanghai datasets. As illustrated, Cross-SPECL demonstrates exceptional feature alignment capabilities. Even without accessing any target data during training, the model produces highly discriminative feature embeddings on the TDs. The distinct semantic categories are well-separated with clear decision boundaries, and the intraclass distributions remain compact. This strong class separability serves as compelling qualitative evidence that Cross-SPECL effectively filters out domain-specific style variations and captures the intrinsic, domain-invariant semantics required for robust cross-scene classification.

G. Time Consumption and Model Complexity

To evaluate the computational efficiency of the proposed method, Table X compares the total trainable parameters and execution time per epoch of Cross-SPECL against seven state-of-the-art methods under identical experimental conditions.

Regarding model complexity, Cross-SPECL demonstrates superior parameter efficiency. It achieves the most compact architecture on both the Houston 2013 (1.03 M) and Hangzhou (1.34 M) datasets, and remains highly competitive on the Pavia University dataset with only 1.14 M parameters. Compared to parameter-heavy DA methods like DSAN (16 M) and LDGnet (36 M), Cross-SPECL is significantly more lightweight, verifying that our framework effectively extracts domain-invariant features without relying on redundant network capacity.

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In terms of training speed, Cross-SPECL consistently achieves the lowest execution time per epoch across all datasets. Notably, compared to DG methods that rely on computationally intensive adversarial training or generative processes (e.g., SDEnet, S^2AMSnet, and FDGNet), Cross-SPECL exhibits a distinct speed advantage. For instance, on the large-scale Hangzhou dataset, Cross-SPECL reduces the training time by approximately 28% compared to the second-fastest method, SDEnet. This efficiency is directly attributed to our cohesive nonadversarial design, which avoids the significant computational overhead associated with training additional generators or discriminators.

VI. Conclusion

In this article, we addressed the challenging problem of single-source DG for cross-scene HSI classification by proposing Cross-SPECL, a cohesive nonadversarial framework. Departing from conventional strategies that rely on unstable adversarial data diversification, our approach establishes a novel "Stabilize-then-Disentangle" paradigm designed to extract strictly domain-invariant semantics. Specifically, we first introduced an SPLT network combined with SFS. By identifying that domain styles primarily manifest in low-frequency components, this module utilizes controlled perturbations and contrastive alignment to neutralize spectral shifts, thereby constructing a robust and representative feature space. Building upon this stabilized foundation, we incorporated a Domain-Agnostic DAG Learning mechanism to explicitly uncover the causal structure of the data. This component generates a causal mask that effectively prunes domain-specific spurious correlations, while CPA further refines the feature space to ensure that the final predictions are driven solely by invariant causal factors. Extensive experiments on three diverse cross-scene benchmarks, including the transfer tasks from Houston 2013 to Houston 2018, Pavia University to Pavia Center, and Hangzhou to Shanghai, demonstrate that Cross-SPECL significantly outperforms state-of-the-art methods. Furthermore, our analysis confirms that by avoiding the computational overhead and instability associated with adversarial training, Cross-SPECL achieves superior generalization performance while maintaining remarkable model compactness and training efficiency.

Despite the promising results, we acknowledge certain limitations in our current framework. A primary challenge lies in the hyperparameter sensitivity associated with the multicomponent architecture. The proposed method involves several key hyperparameters, such as the bandwidth ratio γ for frequency perturbation and the balancing weights λ for the DAG and contrastive losses. While our extensive parameter analysis demonstrates robustness within effective ranges, achieving optimal performance requires careful tuning to balance the tradeoff between feature stabilization and causal structure learning.

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Xi Chen received the B.S. degree in electronic information engineering and the M.S. degree in information and communication engineering from Dalian Maritime University, Dalian, China, in 2020 and 2023, respectively. He is currently pursuing the Ph.D. degree with the College of Systems Engineering, National University of Defense Technology, Changsha, China.

His research interests include hyperspectral image processing and deep learning.



Maojun Zhang received the B.S. and Ph.D. degrees in system engineering from the National University of Defense Technology, Changsha, China, in 1992 and 1997, respectively.

He is currently a Professor with the Department of System Engineering, National University of Defense Technology. His research interests focus on computer vision, information system engineering, and system simulation.



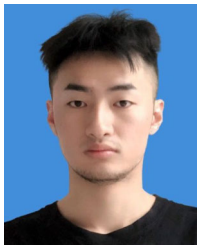
Yuxiang Liu received the B.S. degree in information management and information system from Tianjin University of Technology (TJUT), Tianjin, China, in 2016, and the M.S. degree in control science and engineering from the National University of Defense Technology, Changsha, China, in 2020, where he is currently pursuing the Ph.D. degree.

His research interests include thermal imaging, multispectral matching, and 3-D thermal reconstruction.



Chen Chen received the B.S. degree from Lanzhou University, Lanzhou, China, in 2016, and the Ph.D. degree from the National University of Defense Technology, Changsha, China, in 2021.

He is currently a Research Associate with the College of Systems Engineering, National University of Defense Technology. His research interests focus on object detection and hyperspectral image processing.



Shen Yan received the B.S. and Ph.D. degrees in system engineering from the National University of Defense Technology, Changsha, China, in 2019 and 2023, respectively.

He is currently a Lecturer with the College of Systems Engineering, National University of Defense Technology. His research interest focuses on image-based 3-D reconstruction and computer vision.

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